CROPSENSE AI

Three model signals. One honest demo.

Disease • canopy stress • pest risk

Evaluation, limits, and a laptop-ready walkthrough

01

LEAF RGB

38 disease classes

02

RGB + NIR

NDVI stress classes

03

4-WEEK SERIES

Pest-risk classes

Separate datasets and models; the combined dashboard score is illustrative.

Hyperspectral analysis is outside this demonstration scope.

MONDAY • PROJECT REVIEW

SYSTEM MAP

Three inputs stay distinct

The dashboard brings outputs together for a walkthrough; training data are not a joined farm dataset.

01 LEAF HEALTH

RGB leaf photo

MobileNetV2
38 classes

Disease label + Grad-CAM

02 CANOPY STRESS

RGB + NIR tile

NDVI features
Random Forest

Healthy / moderate / severe

03 PEST RISK

4 weeks of weather
+ prior observations

LSTM sequence
3 classes

Risk class + probabilities

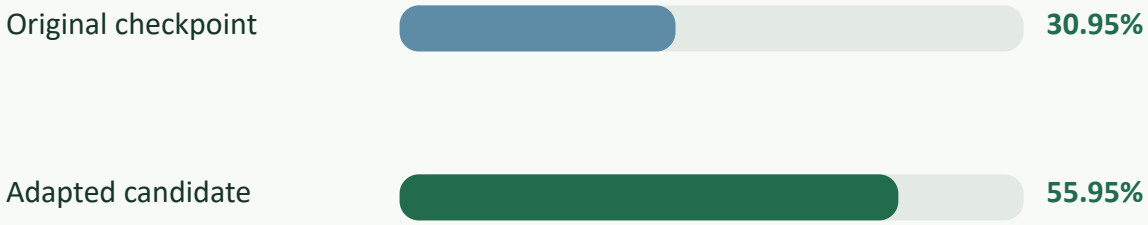
Illustrative composite: equal-weight average of available module scores

No joined ground truth → no measured composite accuracy.

Field adaptation improved the PlantDoc benchmark

The same 252-image field test was used to compare the original checkpoint with the adapted candidate.

PlantDoc test accuracy



141 / 252
adapted correct

0.552
macro F1

Trade-off to state clearly

PlantVillage validation fell from 98.10% to 96.83%.
A small Tomato Late Blight group also lost recall.

PlantDoc test n = 252 • PlantVillage validation n = 8,145

Interpretation: stronger cross-domain result, with a lab-domain and class-level cost.

Grouped field testing exposed the healthy-class weakness

RGB and NIR pixels → NDVI → six summary statistics → three-class Random Forest.

RGB + NIR
paired tile

→

NDVI
per pixel

→

6 features
mean • std
min • max
p25 • p75

→

Random Forest
3 classes

58.42%
grouped test accuracy

0.474
macro F1

23 / 280
healthy recall

74 / 700
severe tiles predicted healthy

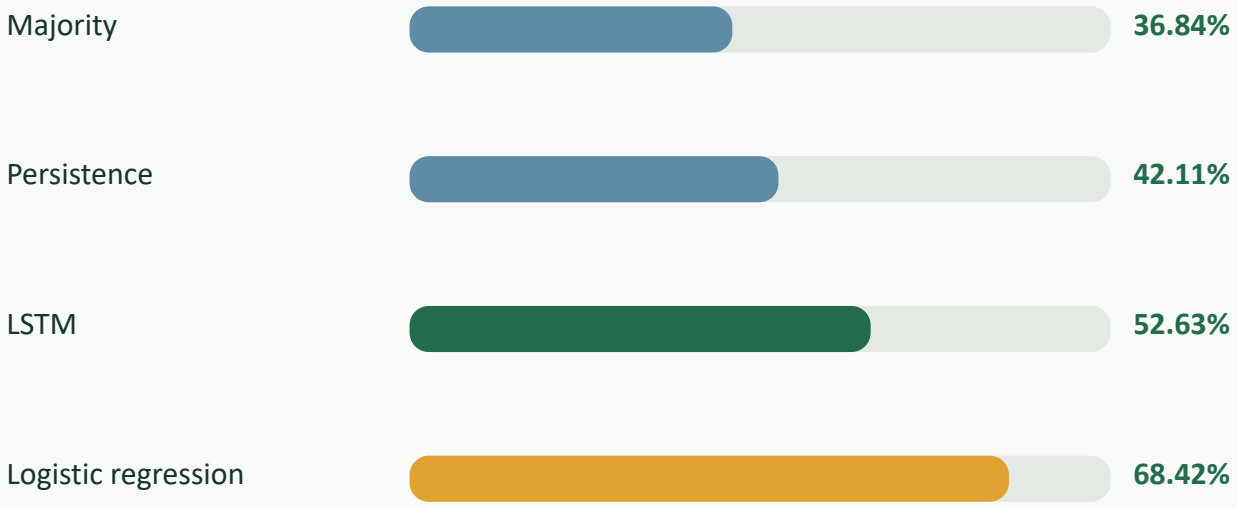
1,549 held-out tiles • split by inferred field ID • deployed older model retained

A threshold candidate raised healthy recall but also raised dangerous severe → healthy errors (74 → 81); rejected.

LSTM beats persistence; logistic regression is stronger

A time-separated, retrospective evaluation on 19 observed target weeks.

Held-out test accuracy



10 / 19

LSTM correct • 52.63%

0.433

LSTM macro F1

Train 2005–07 → validate 2008 → test 2009

Four-week windows • observed targets only • one region • small test set

We kept the final test for measurement, not tuning

Each audit notebook is executed and records enough detail to reproduce the comparison.

DATA

provenance, exclusions, labels

SPLIT

groups or time order

MODEL

configuration, environment, artifact hash

METRICS

accuracy, precision, recall, F1, support

ERRORS

confusion counts + class report

DECISION

selection rule before final test

The dashboard score is not an evaluated model metric. No common farm-level labels across the three source datasets.

Audit notebooks and detailed reports are linked in the project README and docs/MODEL_EVALUATION.md.

Run the same portable app on the Windows 10 laptop

Python 3.12.2 is in scope; the app uses CPU TensorFlow on Windows.

1	**Disease**	PlantVillage leaf → class + Grad-CAM
2	**Canopy stress**	Load paired RGB/NIR sample → NDVI heatmap + stats
3	**Pest risk**	Run a held-out four-week sequence → probabilities
4	**Composite**	Show illustrative score and explain its limits

Laptop setup

1 Clone monday-demo branch
2 Double-click setup_windows.cmd
3 Check environment report
4 Disconnect Wi-Fi; run start_windows.cmd

Laptop rehearsal is still required.

Windows CI passed on Windows Server 2025; the Mac mini also passed. This does not replace the Win 10 laptop check.